Safety Monitor Pro

Complete plugin workflow — install, licensing, runtime, modules, enforcement & updates

WHM / cPanel plugin | Version 1.0.198 | License: AiTech (RSA-signed) | 2026-06-24

★ Module color key

Monitoring · Security · Mail · Ops & Tools · Intelligence · Alerts · Fleet · System

1 Macro lifecycle

Purchase — Storefront → key issued

→ **Install** — One-liner (UA-gated)

→ **Activate** — Key + fingerprint

→ **Bind** — Locked to the server

→

Operate — Daily use · modules

→ **Auto-update** — Signed + pinned

→ **Enforce** — Always-on guard

2 Install & license activation

1. **Customer receives the license key** by email / WhatsApp (auto-sent on serial creation), with the install one-liner.

2. **Runs the installer** — `curl -A "…SafetyMonitorPro/installer" …/install.sh | bash -s -- <KEY>`
   Custom UA bypasses Cloudflare; plain curl is blocked.

3. **Installer computes the server identity** — main IP (`/var/cpanel/mainip`) + hardware fingerprint `sha256(machine-id|hostname)` → **POST /api/download**.

4. **License server validates & binds** — checks key (not expired/suspended/tampered), **binds on first download** (ip+host+fingerprint), enforces single-download, then **streams the obfuscated tarball** + `X-Smpro-SHA256` .

5. **Installer verifies & deploys** — checks SHA, extracts, registers the WHM addon (icon + ACLs + entry URL).

6. **First open → activation** — License page → **POST /api/check** → server returns an **RSA-signed verdict** (status, expiry, features[], bound ip/fp, signed timestamp, feat_key, pair_key). Plugin verifies the signature against the pinned public key & caches it in `state.json` .

3 Runtime architecture

WHM
Admin opens the addon

→

index.cgi
bash wrapper → cPanel PHP

→

index.php
router + boot

→

modules/<page>
the requested feature

Boot gates (every request): defines → **license gate** (is_valid / dormant / tamper-lock) → **integrity gate** (signed manifest + critical-file hashes) → module loader. **Background:** the notify cron runs the license heartbeat (before the gate) + workers (login-guard, cooler, backup-retry, hub, update-check, telemetry/pairing beacon).

4 Module map

Monitoring

- Dashboard
- Server Health
- Load Monitor
- Top Processes
- Disk / Storage
- Network Traffic
- Hardware & Platform
- cPanel Services

Security

- Firewall Status
- Account Audit
- Intrusion / Imunify
- Forensics / Audit
- PHP Harden
- Security Health

Mail

- Spam & Phishing
- Email Queue

Ops & Tools

- Site Doctor
- Server Tune-up
- Database Cleanup
- Backup Monitor
- Analytics / Reports
- Reports Center

Intelligence

- AI Providers (BYO key)

Alerts

- Notifications
- Telegram Bot
- WhatsApp

Fleet

- Multi-Server Hub

System

- License
- Updates
- Plugin Control
- Protection Profiles
- Terms & Legal

Each module is gated by the license's signed `features[]` — a plan that doesn't include a feature hides its page entirely.

5 License enforcement & anti-tamper

Transport outage reversible

License server unreachable → 48h graduated grace (banner escalates, heavy workers pause) → keeps running, auto-recovers on reconnect.

Commercial stop (expired / suspended / revoked) reversible

A real "no" for 48h → whole plugin goes dormant (UI locked, workers stop). NOT uninstalled — renewing clears it automatically.

Tamper (hosts-blackhole + no valid signature) destructive

Immediate silent full-lock (every page) → after 24h → self-uninstall, unless the operator issues a server-side reset. Admin is alerted (WhatsApp + email); a dead-man cron is the backstop.

Pairing beacon (rolling per-license code) reversible · opt-in

A disguised rolling code proves possession of a per-license secret. Sustained mismatch → admin-reasoned reversible stop (self-heals on a valid beacon; operator can Re-pair / rotate the secret).

Root-adversary hardening: grace measured from the server-signed timestamp (not the local clock) · monotonic clock guard (anti-rollback) · in-process integrity manifest (patching one gate doesn't disable protection) · server-side clone-thrash detection. **Honest ceiling:** a self-hosted plugin on customer-root can still be patched at the PHP level — the absolute fix is a bytecode encoder (ionCube).

6 Secure update channel

Update cron
checks daily

→

/api/version
honors per-license pin

→

/api/download
license + fingerprint gated

→

Verify
.sig + SHA + manifest

→

Apply
atomic swap

Releases are published as `smpro-latest.tar.gz` (+ offline RSA `.sig`) with a signed `integrity.json`. A per-license `pin_version` lets the operator stage rollouts (e.g. one canary box) before promoting the fleet. Tampered files → **Protected Mode**.

7 License server (operator side)

Security core

- Offline RSA private key signs every verdict — never on any client.
- Fingerprint-authoritative binding + automatic IP re-bind.
- Single-download per key (reset per row); SHA + UA gates.
- Admin login via email OTP.

Admin panel & automation

- Dashboard · Licenses · Plans · Servers · Releases · Logs · Settings.
- Create license → auto-deliver key + install command (email / WhatsApp).
- Per-license policy: pin version, escalation, pairing, reset, full-stop.
- Provision API for the storefront (provision / renew / revoke / status).